

- 1- Consider a straight, untwisted wing with NACA 0012 sections, aspect ratio $AR = 6$, taper ratio $\lambda = 0.5$ (root chord 2 m, tip chord 1 m, span 9 m), flying at $U_\infty = 50$ m/s and angle of attack $\alpha = 5^\circ$ (zero-lift angle $\alpha_{L=0} = 0^\circ$), in air of density $\rho = 1.225$ kg/m³; its two-dimensional lift-curve slope is $a_0 = 2\pi$ (rad⁻¹). Using a six-term Fourier expansion for the circulation, solve the six simultaneous “lifting-line” equations for the unknown coefficients $\{A_1, \dots, A_6\}$, then compute the total lift coefficient and the induced-drag coefficient. Implement a short MATLAB or Python routine that solves this problem.
- 2- An airplane has a wing area of 200 ft² and a wingspan of 36 ft. The maximum weight is 3000 lb. The airfoil of the wing has a lift curve slope $C_{L\alpha} = a = 0.095$ per degree and a zero-lift angle of attack $\alpha_{L0} = -2^\circ$. $\tau = 0.10$, and the airplane cruises at a velocity of 150 mi/h at standard sea level conditions. The span efficiency factor is $e = 0.78$. Find the angle of attack α and the drag coefficient C_{Di} during cruise.